Machine Learning for Computational Science and Engineering (2026) 2:18
https://doi.org/10.1007/s44379-026-00067-x

RESEARCH

Integrated machine learning and finite element analysis for rail-wheel wear evaluation based on stress and contact mechanics

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Received: 11 February 2026 / Accepted: 14 April 2026
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Abstract

This manuscript presents a novel, comprehensive study focused on predicting the complex interactions between rail wheels and tracks under wear conditions. The study introduces a unique combination of advanced regression techniques — Gaussian Process Regression (GPR), Support Vector Machines (SVM), Decision Tree (DT), Linear Regression (LR), Non-linear Regression (NLR), and Artificial Neural Networks (ANN) — integrated with a Finite Element (FE) model. The FE model, meticulously developed to simulate the intricate dynamics of wheel-rail interactions, allows for a detailed analysis of wear-related parameters, such as contact pressure, stress distribution, and contact area. A key innovation of this work is the integration of multiple machine learning techniques with FE analysis to predict wear evolution in rail-wheel systems. By validating the FE model against established references and generating an extensive dataset, this study provides a rigorous comparison of each regression method's predictive accuracy. The findings offer crucial insights into the strengths and limitations of each technique, enhancing the applicability of predictive modeling for real-world railway maintenance. The combination of FE analysis with machine learning techniques constitutes a significant advancement in railway engineering, offering an efficient and reliable predictive framework for managing wear in rail-wheel interactions, with direct implications for optimizing maintenance strategies.

Keywords Rail wheel · Finite element · Wearing · Prediction · Artificial intelligence

Nomenclature

Symbol	Description
E_i	Young's modulus in direction i, representing the elastic stiffness in tension/compression
G_{ij}	Shear modulus in plane ij, indicating material rigidity against shear deformation
ν_{ij}	Poisson's ratio in direction ij, ratio of transverse strain to axial strain
ρ_k	Density of material k, mass per unit volume
$SOLID185$	ANSYS element type for 3D modeling, used in FEA
T_{170}	Target surface type in contact modeling, FEA
C_{174}	Contact surface type in contact modeling, FEA
MSE	Mean Squared Error, a measure of model prediction error
RMSE	Root Mean Squared Error, square root of MSE
R^2	Coefficient of determination, measure of model fit
$\mathbf{W}$	Weight matrix in ANN
$\mathbf{b}$	Bias vector in ANN
ϕ	Activation function in ANN

1 Introduction

The contact problem between train wheels and rails forms a critical aspect of railway engineering, characterized by the complex interaction of materials, forces, and dynamics. Addressing this challenge requires a multidisciplinary approach, integrating Finite Element Method (FEM) analysis and experimental tests to define and mitigate potential issues. FEM offers a robust computational framework for

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Published online: 28 April 2026

modeling the contact between the wheel and rail, allowing engineers to simulate various loading conditions and assess factors such as stress distribution, contact pressure, and wear patterns.

By leveraging FEM simulations, researchers can explore into the nuances of the contact problem, exploring how factors like wheel profile, track irregularities, and material properties influence performance. Kuminek et al. [12] comprehensively analyzed the primary factors contributing to the degradation of tram turnout frogs. Through numerical simulations, we investigated the distribution of contact stresses within the tram wheel-frog system. Ma et al. [13] investigated the correlation between interface parameters and the precision of contact solutions. It highlights how the selection of incorrect parameters, such as excessively high or low contact stiffness, coarse mesh, or inappropriate combinations thereof, can detrimentally impact the accuracy of wheel-rail interaction solutions. Aalami et al. [1] presented a FEM comprehensive modeling and analysis of a railway system, encompassing components such as wheels, rails, axles, and pads. By employing elastic-plastic materials, mapped meshing techniques, and aligning the rolling motion of the wheel with contemporary international railway systems. Toumi et al. [21] introduced a novel three-dimensional finite element rolling contact model focusing on the enhanced spin effect between the wheel and rail. The authors proposed model is designed to investigate both normal and tangential contact issues, along with analyzing creep force characteristics in both elasticity and elasto-plasticity conditions.

Experimental analyses are also conducted by research community, such as wheel-rail interface tests conducted on scaled-down models or full-scale track testing, provide invaluable insights into the actual contact behaviour, allowing engineers to refine their FEM models and validate their assumptions. Numerous studies have employed this integrated approach, such as those by [2]. The authors outlined the methodology employed to construct a scaled test bench for experimental assessment of the wheel-rail contact phenomenon.

The study focuses on analysing the longitudinal contact force variation in relation to longitudinal creepage. Marshall et al. [14] examined the evolution of wheel-rail contact patches through experimentation. Three distinct contact scenarios were investigated, including those arising from unused, worn, and sand-damaged wheel and rail specimens. Contact pressure distributions obtained from measurements were compared with those derived from elastic analytical and numerical elastic-plastic solutions.

Wear in wheel-rail systems refers to the gradual loss of material from the contacting surfaces of both the wheel and the rail due to friction and mechanical interactions. The literature [6, 9, 22] highlights several critical factors influencing the rate of rail wear. These factors encompass track curvature, layout, rail grade, metallurgical properties, profile, train speed, wheel profile, axle load, traffic patterns, lubrication practices, the presence of contaminants, and various environmental conditions. This process can result in changes to the surface profiles, affecting the overall performance and longevity of the wheel-rail system. Wear is a critical consideration in railway engineering as it directly impacts maintenance requirements and the efficiency of rail transportation systems. Chongyi et al. [4] proposed and implemented a methodology for predicting wheel/rail profile wear, specifically focusing on wear of the flange and gauge. The simulation of the wheel/rail disc rolling contact process is conducted using the three-dimensional nonlinear finite element dynamic analysis code ABAQUS. Famurewa et al. [7] conducted an analysis of rail profile measurement data spanning from 2008 to 2013. The authors objective was to assess the wear characteristics of curves with varying radii along the Swedish iron ore line. The findings aim to offer valuable engineering insights to enhance maintenance practices for improved rail performance. Tao et al. [20] examined the wear and rolling contact fatigue of wheel-rail materials by manipulating different wheel/rail hardness ratios and creepages. The authors findings revealed that as the wheel/rail hardness ratio increased from 0.927 to 1.218, there was a notable decrease in the wheel wear rate. Cavazzuti [3] introduced a model for assessing the evolution of wheel and rail profiles resulting from wear. This model consists of two interconnected yet distinct components: a vehicle model for dynamic analysis and a separate model for wear estimation. Shebani and Iwnicki [16] analysed wheel and rail wear employing an artificial neural network. Specifically, Nonlinear Autoregressive models with exogenous input neural network (NARXNN) were constructed to predict wheel and rail wear.

The interaction between train wheels and rails is a fundamental issue in railway engineering. Recent advances in machine learning, finite element analysis (FEA), and contact mechanics have provided new insights into wear mechanisms and predictive maintenance strategies ([17, 23]). Machine learning, in particular, has shown promise in predicting wear patterns, reducing computational costs, and enhancing the accuracy of finite element simulations ([17, 19]).

Finite element analysis continues to be central to the study of wheel-rail interactions, particularly in wear prediction and stress distribution analysis. A recent study integrated machine learning with FEA to improve the accuracy of contact area and stress distribution predictions [25]. Furthermore, Bayesian optimization has been employed in recent works to streamline FEA models, achieving significant improvements in computational efficiency [24]. The integration of machine learning algorithms with advanced computational modeling has recently demonstrated significant potential across various complex engineering domains, reinforcing the shift towards data-driven surrogate models.

For instance, recent studies have successfully combined Computational Fluid Dynamics (CFD) with machine learning to accurately estimate heat transfer characteristics and windage losses in electrical machines, significantly reducing computational costs while maintaining high-fidelity predictions [10]. Similarly, in geotechnical engineering, hybrid intelligent modeling and machine learning techniques have been effectively applied to the stochastic reliability modeling of reinforced soil foundations, providing robust predictive frameworks to handle intrinsic material uncertainties [15]. These multidisciplinary advancements underscore the growing capability of AI-based surrogate models to complement and optimize traditional numerical methods.

Several researchers have explored the effects of wear on wheel-rail interactions under dynamic loads. For instance, a 2021 study developed a neural network model that predicts wheel wear under various operational conditions, demonstrating the benefits of machine learning in real-time predictive maintenance [18]. Additionally, advancements in material science have contributed to more robust models, accounting for factors like material hardness and environmental conditions, which are crucial in accurate wear modeling [8].

The purpose of this paper is to develop a computationally efficient surrogate model for rail-wheel interaction based on Finite Element Analysis (FEA) data. Surrogate models have been widely adopted in the railway industry and other engineering fields to reduce computational costs while maintaining sufficient accuracy. These models can approximate complex systems, such as wheel-rail contact mechanics, where full-scale simulations are impractical for real-time applications or extensive parameter studies. Previous works in the literature have explored various types of surrogate models, including those based on polynomial regression, neural networks, and Gaussian processes, each offering different advantages in terms of accuracy and computational speed. This study aims to address the challenge of wear-related evaluation in rail transport by leveraging machine learning to create a surrogate model that significantly reduces the computational burden of FEA while preserving its accuracy. By focusing on specific wear-related parameters, this model offers a practical tool for railway engineers to optimize maintenance strategies without the need for extensive computational resources.

The prediction of wheel-rail wear involves not only accurate modeling of contact mechanics but also the inclusion of material properties and wear mechanisms. While FEA is a powerful tool for simulating the wheel-rail contact mechanics—enabling detailed analysis of contact pressure, stress distributions, and deformation patterns—it does not inherently account for the wear behavior of the materials involved. Predicting wear requires integrating material-specific data, such as hardness, frictional properties, and wear coefficients, which can vary significantly based on environmental conditions and loading histories. One of the challenges in railway applications is that the material properties of rails and wheels are not always well-characterized, and this uncertainty must be managed carefully in wear prediction models. By combining FEA results with empirical wear laws and material-specific inputs, a more comprehensive and accurate wear prediction can be achieved, contributing to optimized maintenance strategies.

In summary, this paper aims to develop a reliable predictive model based on regression algorithms to estimate the contact pressure, stress, and contact area between the rail and wheel, particularly under different wearing profiles. The architecture of this document methodically unfolds the formulation and validation phases of our Finite Element (FE) model, meticulously designed to encapsulate the complex dynamics inherent in Wheel/Rail (W/R) interaction scenarios. Commencing with a comprehensive overview of the FE model, the discourse transitions into an exploration of the methodology employed to parameterize alterations in wear profiles, offering an innovative perspective on the evolution of wear phenomena. Following this, an elaborate exposition of regression algorithms framework is presented, illustrating its capacity to adeptly navigate the complexities associated with predictive modeling tasks. The antepenultimate section is devoted to an exhaustive validation of the FE model, thereby laying the groundwork for a thorough assessment of the predictive model's efficacy. The ensuing discussion not only corroborates the precision and dependability of the proposed framework but also highlights its groundbreaking contribution to the enhancement of predictive maintenance protocols within the realm of railway engineering. By synergizing advanced FE analysis with the predictive acumen of neural networks, this manuscript heralds a seminal advancement in the proactive oversight of rail-wheel interactions, signifying a new epoch of optimization and safety in rail transport systems.

2 FE model of The wheel rail interaction

This section presents the Finite Element (FE) model developed to simulate the wheel–rail interaction using ANSYS APDL [5]. To reduce computational cost while maintaining accuracy in the contact region, a sub-modelling strategy is adopted. Only a localized of the wheel and rail is modeled, as shown in Fig. 1, allowing detailed analysis of the contact mechanics without simulating the full system.

The FE discretization is performed using SOLID185 elements, which are three-dimensional elements with eight nodes and three translational degrees of freedom per node. Contact between the wheel and rail is modeled using surface-to-surface contact pairs, consisting of TARGET170 elements

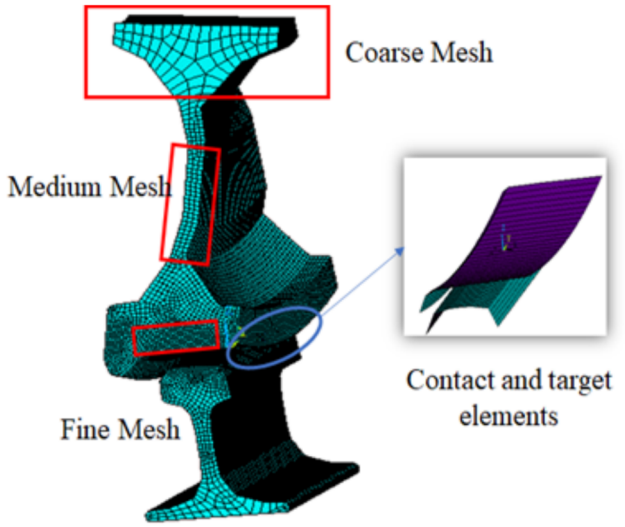


Fig. 1 Finite Element Model

for the target surface and CONTACT174 elements for the contact surface. This formulation enables an accurate representation of nonlinear contact behavior, including pressure distribution and potential sliding.

To further optimize computational efficiency, a non-uniform meshing strategy is employed. The model is divided into three regions characterized by coarse, medium, and fine mesh densities. A refined mesh is applied in the contact to accurately capture stress gradients and contact pressures, while a medium mesh is used in the surrounding. Coarser elements are assigned to regions from the contact zone, where stress variations are less significant. Following a mesh sensitivity analysis, a total of 71,027 elements is selected as a compromise between computational efficiency and solution accuracy. The present model represents a localized finite element analysis of the wheel–rail contact problem. The global vehicle–track dynamic system is not explicitly modeled; instead, a simplified dynamic representation is adopted. Inertia effects are incorporated through the material density and the use of a lumped mass element (MASS21) applied at the wheel reference node. The model does not explicitly include unsprung masses (e.g., wheelset or bogie) or suspended vehicle masses. Furthermore, the track structure (rail pads, sleepers, and ballast) is not modeled as a deformable system; instead, the rail base is fully constrained to approximate support conditions.

While this approach enables a detailed investigation of local contact stresses and pressure distributions, it does not capture the full dynamic coupling between the vehicle and track. Therefore, the results should be interpreted within the context of a simplified global dynamic representation.

2.1 Physical modeling assumptions

The wheel–rail interaction is modeled under simplified but representative operating conditions to balance physical realism and computational efficiency. A vertical load of 300 kN is applied to the wheel to represent a typical axle load in heavy railway systems. The contact between the wheel and rail is modeled using a surface-to-surface contact formulation, allowing for nonlinear contact pressure distribution. The rail is constrained at its base to simulate the support provided by sleepers and ballast. This support is approximated as a rigid foundation, which neglects local flexibility effects but provides a stable boundary condition for contact analysis. The material behavior of both wheel and rail is assumed to be linear elastic, characterized by Young's modulus and Poisson's ratio. While elastoplastic effects are known to influence wear mechanisms, their inclusion would significantly increase computational cost. Therefore, this study focuses on elastic contact behavior as a first-order approximation.Rolling contact is simulated under quasi-static conditions. Creepage effects are not explicitly modeled using a dedicated law (e.g., Kalker theory), but their influence is partially captured through the relative motion and contact formulation.

It is important to note that the full vehicle–track system, including bogie mass, suspension stiffness, and ballast compliance, is not explicitly modeled. Instead, their combined effects are approximated through the applied loading and boundary conditions. This represents a limitation of the current model but allows efficient parametric analysis. Overall, the adopted assumptions enable the generation of consistent and computationally efficient datasets for subsequent machine learning analysis, while maintaining the essential physics of wheel–rail contact.

3 Results and discussion

In this section, we unveil the primary discoveries of our current paper. The initial subsection will provide a brief validation of the proposed Finite Element Model. Following that, a comprehensive presentation of detailed results and crucial insights from the predictive analysis of wheel and rail wear investigation will be expounded upon.

3.1 Validation

To validate the developed Finite Element (FE) model, a comparison was performed with the wheel-rail submodel proposed by Toumi et al. [21]. Following their approach, the upper portion of the wheel and the lower section of the rail were excluded from the three-dimensional model. This

simplification reduces the total number of elements and, consequently, decreases computational cost without significantly affecting the accuracy of the contact analysis. The resulting model configuration is presented in Fig. 2.

The wheel and rail were assumed to be made of R900A steel, commonly used in railway applications. The material properties adopted in this study include a Young's modulus of 210 GPa, a Poisson's ratio of 0.28, and a density of 7800 kg/m^3.

A dynamic rolling simulation was performed using an implicit Newmark-beta transient scheme, in which the wheel undergoes rotation about its axis, as illustrated in Fig. 2. A time step of 3.7×10^{-4} s was selected based on the ratio between the contact patch size and the rolling velocity, ensuring both numerical stability and adequate temporal resolution of the contact mechanics.

A comparison of the contact pressure distribution and the corresponding contact surface area with those reported by Toumi et al. [21] is presented in Fig. 3. In addition, the contour of the contact shear stress is also compared. The reference model predicts a maximum contact shear stress of 210 MPa, whereas the present FE model yields a slightly higher value of 240 MPa, corresponding to a deviation of approximately 14.29%, as summarized in Table 1.

Furthermore, the distribution of contact pressure along the rolling direction (x-axis) is plotted and compared with the results of Toumi et al., as shown in Fig. 4. For validation purposes, both static and dynamic analysis schemes were employed. The static analysis was carried out using a nonlinear Newton–Raphson iterative method, which neglects inertial effects and is suitable for steady-state conditions. This approach ensures an accurate representation of material nonlinearity and load–deformation behavior.

In contrast, the dynamic analysis was performed using the implicit Newmark-β time integration scheme, which accounts for inertial effects and is therefore more appropriate for transient loading conditions. The results indicate that the dynamic model provides improved accuracy, predicting a maximum normal contact stress of 1118 MPa compared to 1108 MPa obtained from the static model. When compared with the reference value of 1126 MPa reported by Toumi et al. [21], the dynamic model shows a very close agreement with a minimal deviation of only 0.71%, whereas the static model exhibits a slightly larger deviation of 1.60% (see Table 1).

These findings demonstrate that the dynamic formulation, by incorporating inertia and time-dependent effects, offers a more realistic representation of wheel–rail contact behavior.

3.2 Wearing parametrisation

In the Finite Element (FE) model, we systematically address wear in both the wheel and rail by strategically modifying their profiles. This adjustment involves parametric tweaks, where we manipulate the coordinates of specific control key points that serve as pivotal locations guiding the geometric profile changes. The parametric adjustments occur within a defined interval, ranging from 0 to 10 millimeters. This interval allows for a thorough exploration of how geometric profile changes affect contact pressure, contact area, and stress, offering valuable insights into the dynamic behavior of the wheel-rail interaction. To effectively map this complex design space, a stochastic approach was strictly necessary and thus employed using five control points. Specifically, a uniform random sampling strategy was utilized to generate 1500 independent geometric configurations. For each of the 1500 iterations, five random values uniformly distributed between 0 and 10 mm were computationally generated to represent the spatial displacement of the control points. These randomized geometric arrays were then fed directly into the ANSYS APDL code as parametric inputs to simulate the complex profile variations of the rail and wheel. Figure 7 visually depicts the critical control key points used to manipulate both profiles in our study.

In this context, we are working with five control points that are randomly selected over a total of 1500 iterations.

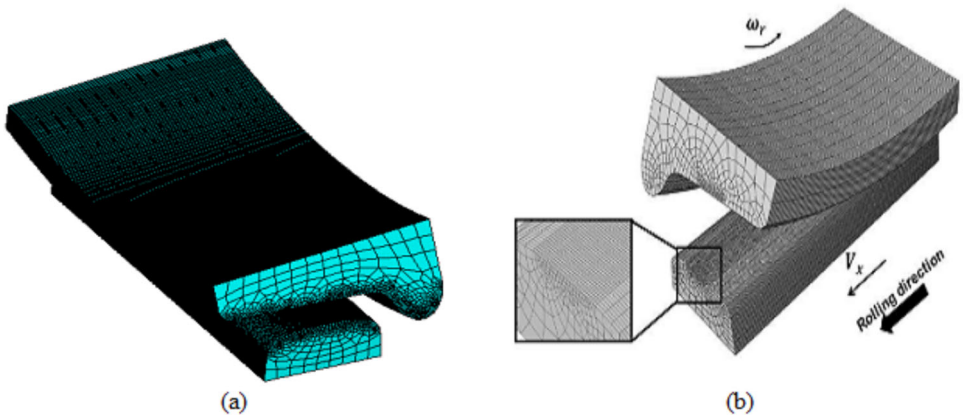


Fig. 2 Finite element model: (**a**) present study and (**b**) reference [21]

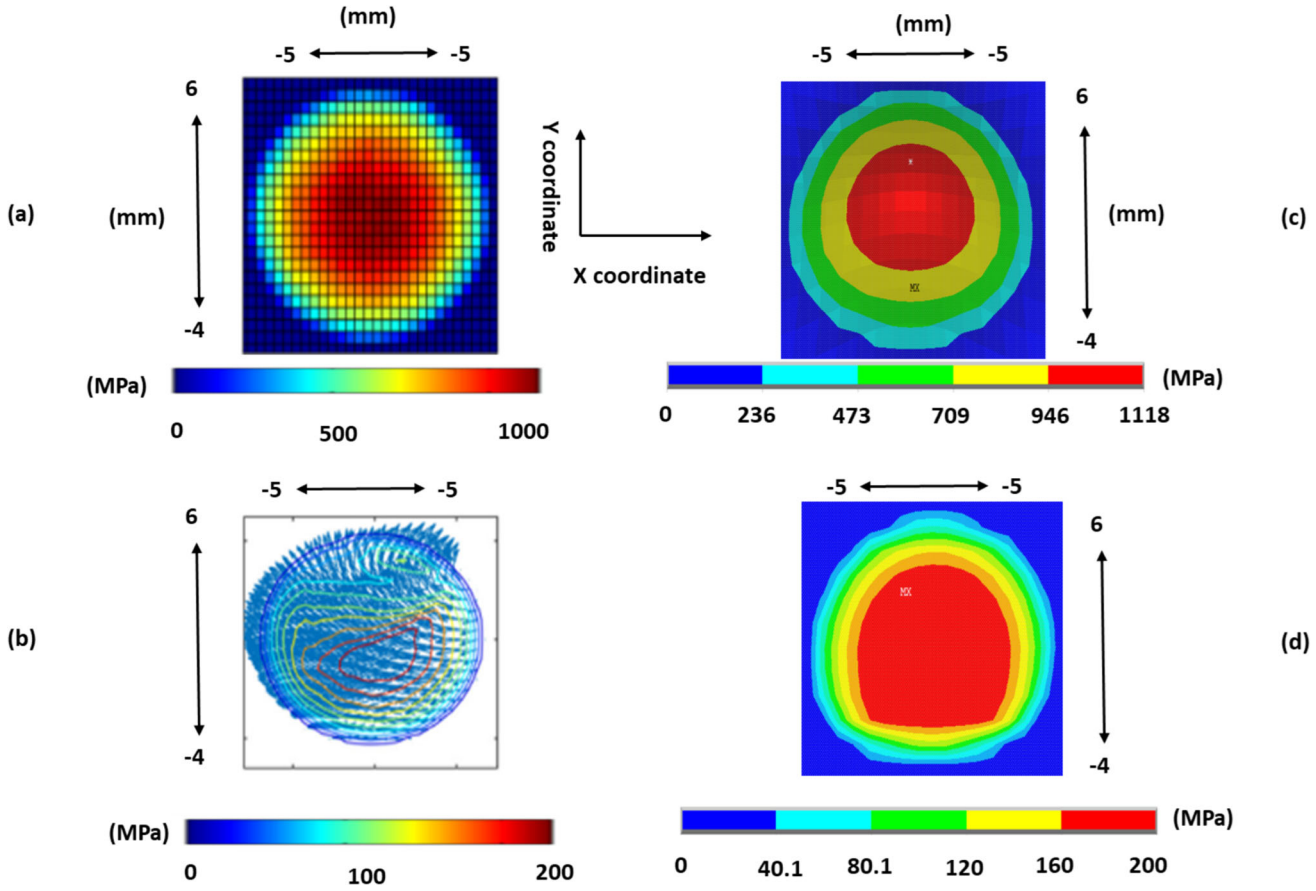


Fig. 3 Wheel–rail implicit dynamic contact results: (**a**) normal contact pressure [21], (**b**) normal contact pressure (present study), (**c**) contact shear stress [21], and (**d**) contact shear stress (present study)

The variable `N_Run` is set to 1500, representing the number of iterations. A MATLAB command, `x = rand(5, N_Run) * 10e-3`, is used to generate a matrix `x` of random values, where the function `rand(5, N_Run)` creates a 5×1500 matrix of uniformly distributed random numbers between 0 and 1. These values are then scaled by multiplying by 10^{-2} (i.e., 10×10^{-3}), resulting in entries that range from 0 to 0.01. For each of the 1500 runs, five random values are generated within this range, representing the control points. Each column of this matrix corresponds to a single simulation run, defining a unique set of five control point variations. These five control points, represented by the rows of the matrix, are used as input to ANSYS APDL code to simulate the profile variation of the rail and wheel.

Figure 5 visually depicts the critical control key points used to manipulate both the wheel and rail profiles in our study.

To generate the dataset for the predictive model, the wheel–rail finite element model is implemented as described in Section 2. Following a series of mesh sensitivity analyses, a discretization consisting of 67,912 elements is adopted, ensuring a balance between computational efficiency and accuracy in capturing contact stresses.

The simulation is carried out using a two-step procedure. In the first step, a quasi-static preload is applied to establish stable wheel–rail contact. In the second step, a fully transient dynamic analysis is performed using the implicit Newmark-β integration scheme [11]. The load acting on the

Table 1 Comparison of FE results with Toumi et al. [21]

Parameter	Toumi et al.	Present	Error (%)
Contact Shear Stress (MPa)	210	240	+14.29%
Normal Contact Stress - Static (MPa)	1126	1108	-1.60%
Normal Contact Stress - Dynamic (MPa)	1126	1118	-0.71%

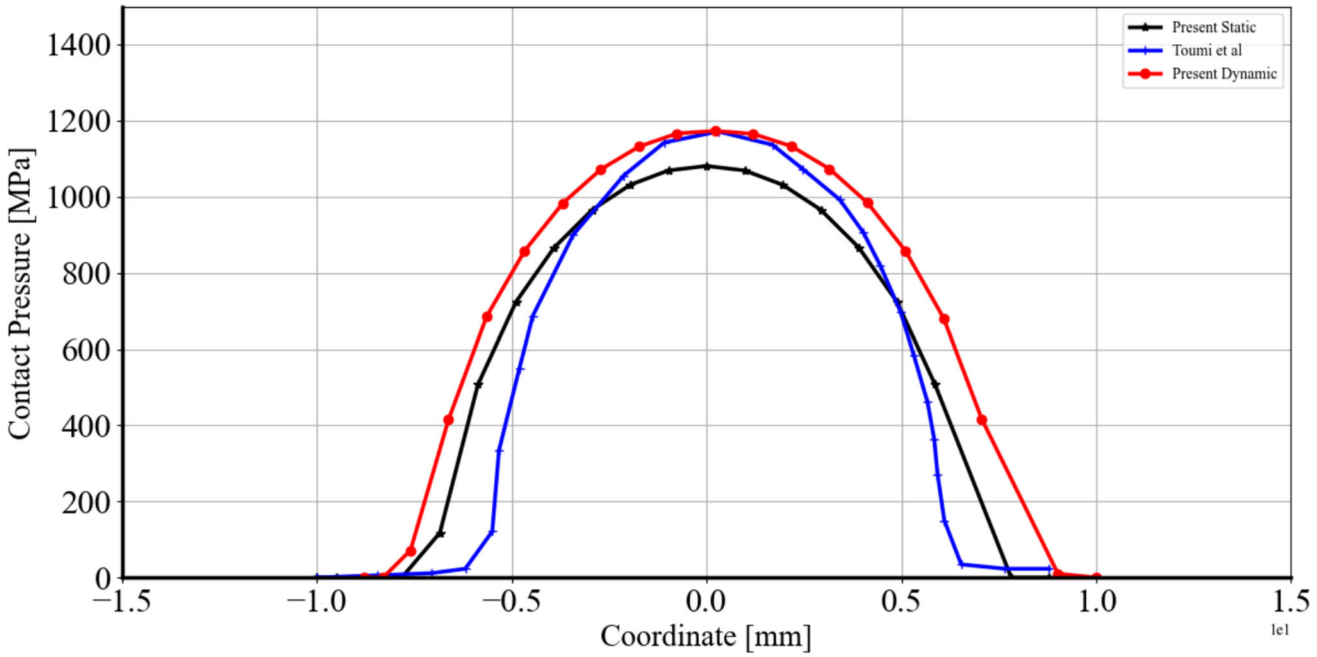


Fig. 4 Contact pressure distribution along the x-direction

system is applied as a vertical nodal force in the z-direction (300 kN), while the rolling motion of the wheel is imposed through prescribed translational and angular velocities of −40.27 m/s and 93.40 rad/s, respectively. A time increment of $\Delta t = 3.7 \times 10^{-4}$ s is adopted. During the preload phase, this value corresponds to the load increment required to ensure convergence of the nonlinear contact problem and to avoid excessive penetration at the interface. During the transient phase, automatic time stepping is employed to maintain

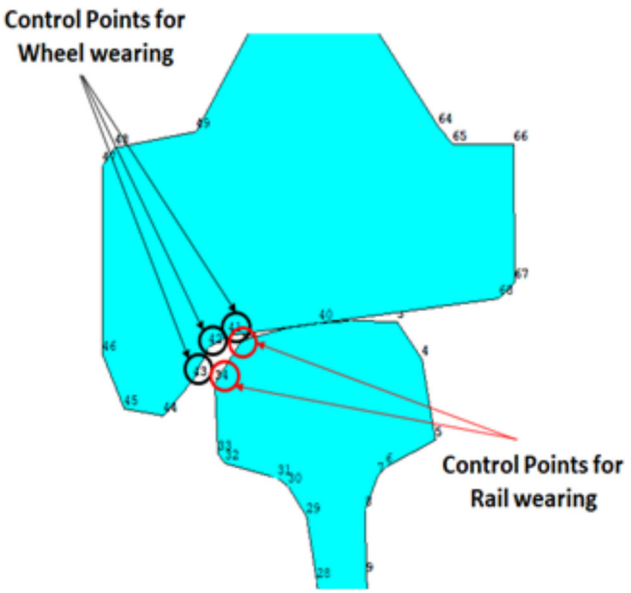


Fig. 5 Wearing profile control points

numerical stability and accuracy. The characteristic time step was selected based on the ratio between the contact patch size and rolling velocity, ensuring adequate temporal resolution of the contact mechanics. We conducted 1500 iterations, integrating five design variables as previously mentioned. Among these variables, two control points are designated for the rail wearing profile, labeled RailA and RailB, while the remaining three control points are allocated to the wheel wearing profile, identified as WheelA, WheelB, and WheelC. Ten examples of the wearing profiles are depicted and presented in Fig. 6.

Figure 7 shows, as an example, some designs obtained depending on different parameters of the design vector X. Additionally, Fig. 8 shows the contact pressure for the first 4 designs. It is observed that even a small variation of a few millimeters or tenths of millimeters is capable of significantly altering the contact pressure.

In the simulations presented in Fig. 7, a correct mesh size was deliberately used in specific regions of the model to balance computational efficiency and accuracy. The simulations focused on capturing the essential contact mechanics between the wheel and rail, and in regions away from the contact interface, a lower mesh resolution was sufficient. To ensure accuracy where it is most critical, such as in the contact zones where stress and pressure distributions are the primary focus, a finer mesh was applied. This adaptive meshing strategy significantly reduces computation time without compromising the accuracy of the simulation results in the key areas. The contact region's finer mesh ensures that the complexities of non-Hertzian contact behavior

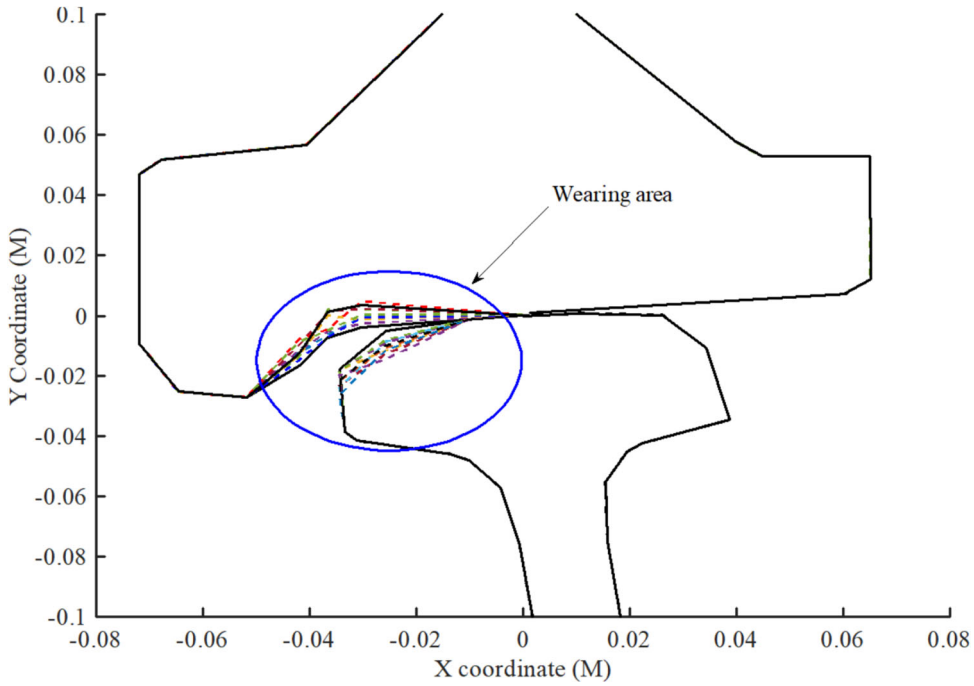


Fig. 6 Examples of collected wearing profiles

are well-represented, while less computationally intensive coarse mesh regions are used elsewhere in the model.

Finite Element Analysis (FEA) models, especially those simulating complex non-linear interactions like wheel-rail contacts, demand significant computational resources. In this study, the computational time for a single simulation varied depending on the specific profile configuration, taking approximately a couple of minutes to complete on an Intel Core i7 processor equipped with 64 GB of RAM. To generate the extensive dataset of 1500 samples required for training the machine learning models within a feasible timeframe, batch processing techniques were utilized. Instead of true parallel computing—which divides a single problem across multiple cores—parametric sweeping was employed to execute multiple independent FEA simulations concurrently. This approach significantly accelerated the data generation process, ensuring that a comprehensive dataset could be assembled to effectively train and validate the AI surrogate models.

To eliminate any ambiguity regarding the data generation process, it is important to emphasize that the entire dataset consists of exactly 1500 unique samples, all of which were obtained directly from the full dynamic Finite Element Analysis (FEA) simulations. No interpolation, artificial smoothing, or data augmentation techniques were employed to expand the dataset beyond these actual FE runs. Consequently, both the training (60%) and validation (40%) subsets are comprised exclusively of true physical simulation data. This ensures that the evaluated machine learning performance metrics accurately reflect the models' genuine predictive capabilities without any risk of data leakage.

3.3 Predictive model

This subsection elaborates on the predictive models utilized to forecast wear in rail-wheel interactions based on various inputs and outputs. The regression models include Gaussian Process Regression (GPR), Support Vector Machines (SVM), Decision Trees (DT), Linear Regression (LR), Nonlinear Regression (NLR), and Artificial Neural Networks (ANN). Each model is briefly introduced with its theoretical foundation and equations, followed by the methodology employed for their application in this study.

To ensure optimal predictive performance, a rigorous hyperparameter tuning procedure was implemented for the regression models. For the Gaussian Process Regression (GPR), Support Vector Machines (SVM), Decision Trees (DT), and Linear Regression (LR), an automated Bayesian Optimization routine was employed. This probabilistic model-based optimization utilized the expected-improvement-plus acquisition function to efficiently explore the hyperparameter space and automatically select the optimal kernel scales, box constraints, and tree depths that minimized the cross-validation loss. The Non-Linear Regression (NLR) was implemented as a deterministic, full quadratic Response Surface Model (RSM) encompassing all linear, interaction, and squared terms of the five inputs.

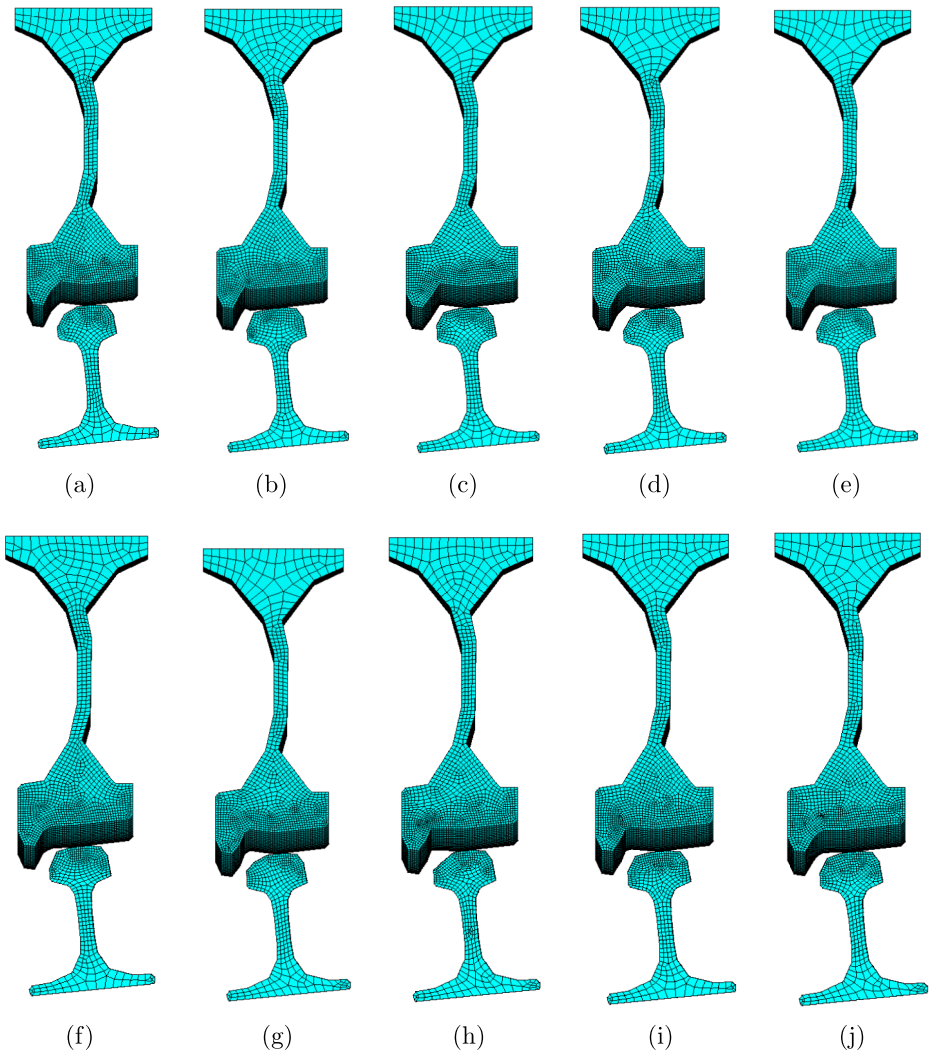


Fig. 7 Some examples of different configurations from the dataset

Gaussian Process Regression (GPR) is a non-parametric, Bayesian approach to regression. It is particularly useful for estimating the uncertainty of predictions. The foundation of GPR is the Gaussian process, defined by the Eq. 1.

$$f(x) \sim \mathcal{GP}(m(x), k(x, x')), \tag{1}$$

where $m(x)$ is the mean function, and $k(x, x')$ is the covariance function between points x and x'.

Support Vector Machines (SVM) are primarily used for classification tasks but can be adapted for regression, known as Support Vector Regression (SVR). The objective is to find a function $f(x)$ that deviates from the actual observed outputs y by a value no greater than ϵ for each training point, and at the same time, is as flat as possible. In linear form, it is represented as shown in Eq. 2.

$$f(x) = \mathbf{w}^T \mathbf{x} + b, \tag{2}$$

where $\mathbf{w}$ is the weight vector, and b is the bias.

Decision Trees (DT) are a non-parametric supervised learning method used for classification and regression. The goal is to create a model that predicts the value of a target variable by learning simple decision rules inferred from the data features. For a regression tree, the model predicts the target by learning the decision rules that minimize the mean squared error (MSE).

Linear Regression (LR) is a basic yet powerful tool for regression analysis, estimating the relationship between one or more independent variables and a dependent variable. The linear regression model can be expressed as shown in Eq. 3

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n + \epsilon, \tag{3}$$

where β_0 is the intercept, $\beta_1, \ldots, \beta_n$ are the coefficients, and ϵ is the error term.

Non-linear Regression (NLR) models the relationship between a set of predictor variables and a response variable

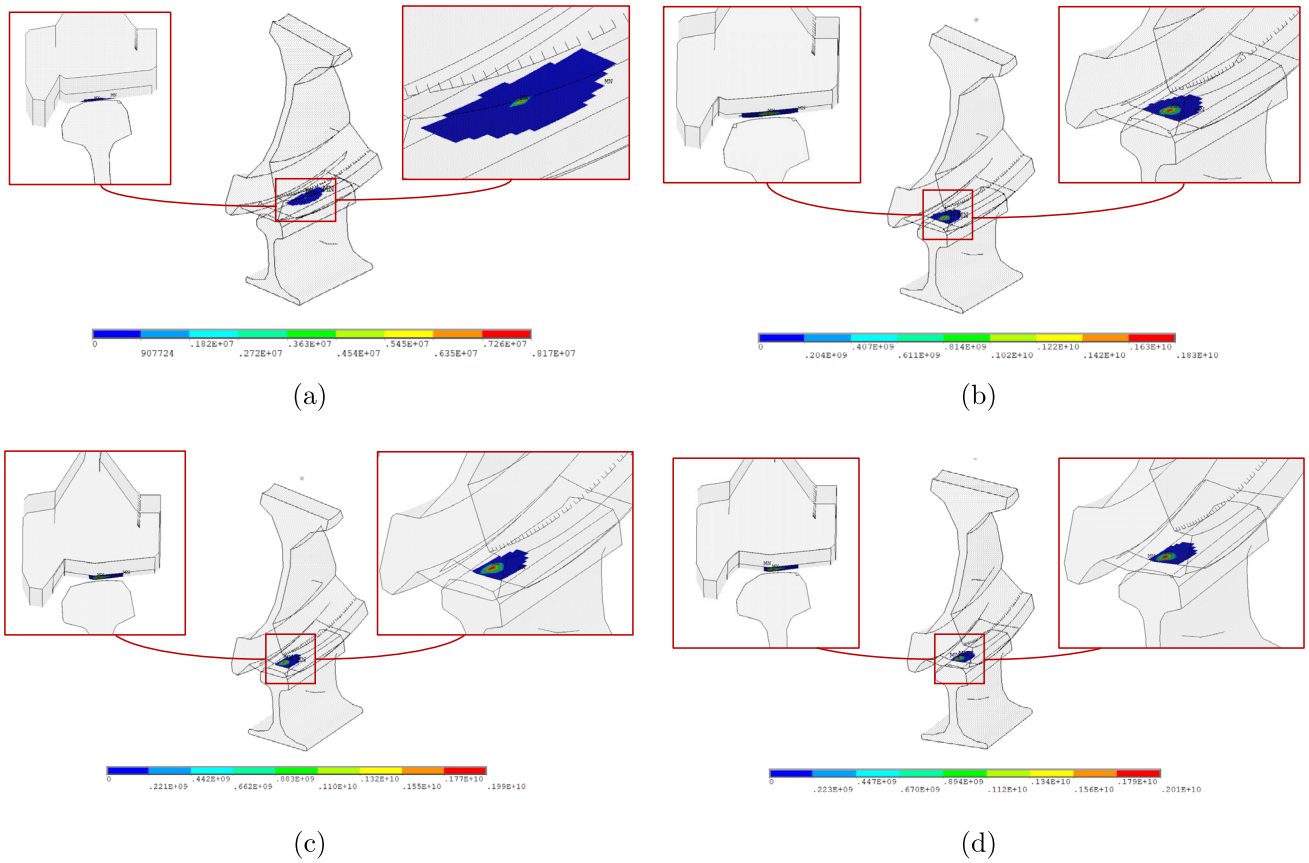


Fig. 8 Contact pressure results for some examples of different configurations from the dataset

by a non-linear equation (Eq. 4).

$$y = f(\mathbf{x}, \boldsymbol{\beta}) + \epsilon, \tag{4}$$

where f is a non-linear function, $\mathbf{x}$ is the vector of predictors, $\boldsymbol{\beta}$ is the vector of coefficients, and ϵ is the error term.

Artificial Neural Networks (ANN) are computing systems inspired by the biological neural networks. In the context of regression, an ANN can be represented by a series of layers, including an input layer, hidden layers, and an output layer. The relationship between inputs and outputs in a network with a single hidden layer can be described by Eq. 5.

$$y = \phi(\mathbf{W}_2 \cdot \phi(\mathbf{W}_1 \cdot \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2), \tag{5}$$

where $\mathbf{W}_1, \mathbf{W}_2$ are the weight matrices, $\mathbf{b}_1, \mathbf{b}_2$ are the bias vectors, and ϕ represents the activation function.

For the Artificial Neural Network (ANN), prior to training, both the input features and target outputs were standardized to a zero mean and unit variance (Z-score normalization) to ensure numerical stability. The network architecture was specifically configured as a feed-forward network comprising a single hidden layer with 40 neurons. The hyperbolic tangent sigmoid function (`tansig`) was applied as the activation function in the hidden layer, while a linear transfer function (`purelin`) was utilized in the output layer. The network was trained using the robust Levenberg-Marquardt backpropagation algorithm (`trainlm`). The training hyperparameters were explicitly set with a learning rate of 0.15, a minimum performance gradient of 1e-20, and a maximum of 10,000 epochs, providing a highly capable architecture to map the complex non-linearities of the contact mechanics.

The complete dataset, comprising 1500 deterministic samples with geometric inputs (Rail A, Rail B, Wheel A, Wheel B, Wheel C) and corresponding mechanical outputs (contact area, maximum pressure, maximum stress), was randomly partitioned into a training set and a testing/validation set. Specifically, 60% of the data (900 samples) was allocated for training the algorithms and optimizing the hyperparameters, while the remaining 40% (600 samples) was retained as a strict hold-out testing set to evaluate the models' generalization capabilities on unseen configurations. Due to the deterministic nature of the computational simulations, a distinct third subset for validation was not required, as the 600 testing samples provided a statistically substantial envelope to robustly measure the surrogate models' predictive accuracy.

It is important to highlight the rationale behind the chosen 60-40 train-validation split. While K-fold cross-validation

is widely recognized for robust generalization assessment, the 60-40 hold-out method was deemed highly adequate for this study due to the deterministic nature of the Finite Element (FE) generated data. Unlike experimental datasets characterized by random noise, the FEA outputs are deterministic responses to uniformly sampled parametric inputs. Consequently, the 40% validation subset yields an absolute size of 600 strictly unseen configurations, providing a statistically substantial and comprehensive representation of the design space to test model generalization. Furthermore, integrating a K-fold cross-validation scheme alongside the computationally intensive Bayesian Optimization process for hyperparameter tuning—across six different mathematical architectures—would be computationally prohibitive. Thus, the large fixed validation set successfully balances rigorous model evaluation with computational feasibility.

The selection of regression techniques encompassed both linear and non-linear models to capture the potential complexity of the rail-wheel interaction phenomena under study. The performance of each model was evaluated based on standard metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) values to quantify the prediction accuracy and model fitness.

To ensure a fair comparison and thorough understanding of each model's strengths and weaknesses, cross-validation techniques were also employed where applicable. This approach helped in mitigating overfitting and provided a more generalized performance estimation across different subsets of the dataset.

Prior to training, the dataset underwent several preprocessing steps, including normalization of the input features to a common scale, handling of missing values, if any, and a preliminary analysis to identify potential outliers. Such preprocessing steps were crucial to ensure that the models trained on this data could learn effectively and provide reliable predictions.

Each model's hyperparameters were carefully selected and tuned using a combination of grid search and cross-validation techniques. This iterative process involved training multiple models with different hyperparameter configurations and selecting the one that performed best on a validation set or through cross-validation.

After training, each model was evaluated on the validation set. This step involved computing the prediction errors and comparing the predicted outputs against the actual values to assess the model's performance. The evaluation phase was critical for identifying the most effective regression technique(s) for predicting wear in rail-wheel interactions based on the predefined metrics.

The performance of each regression model was then analyzed and discussed, highlighting their predictive capabilities, advantages, and limitations in the context of wear prediction in rail-wheel interactions. Insights gained from this comparative analysis informed the selection of the most suitable model(s) for further development and application in railway engineering practices.

Table 2 presents the comprehensive numerical comparison of all evaluated machine learning models based on MSE, RMSE, and R^2. The inclusion of these quantitative metrics allows for an objective assessment beyond visual surface plots, revealing nuanced behaviors among the algorithms.

The analysis demonstrates that the Decision Tree (DT) algorithm is the most robust and consistent surrogate model across all three output parameters. The DT model successfully captured the highly non-linear relationships, achieving remarkable accuracy for Contact Area (R^2 = 0.959), Maximum Pressure (R^2 = 0.974), and Maximum Stress (R^2 = 0.829). In contrast, while the Artificial Neural Network (ANN) outperformed all other algorithms in predicting Maximum Pressure (R^2 = 0.982), it struggled significantly to generalize the Contact Area and Maximum Stress, resulting in negative R^2 values. This indicates that the single-architecture ANN utilized in this study may suffer from overfitting or scale-sensitivity when mapping multiple mechanical responses simultaneously. Furthermore, Linear Regression (LR) and Support Vector Machines (SVM) consistently underperformed, confirming that simple linear or

Table 2 Performance metrics (MSE, RMSE, and R^2) for the evaluated regression algorithms across all three target variables

Algorithm	MSE	RMSE	R^2
Contact Area (mm^2)			
GPR	1.3825e-08	1.1758e-04	-0.00516
SVM	1.3825e-08	1.1758e-04	-0.00516
DT	5.5125e-10	2.3479e-05	0.95992
LR	1.3825e-08	1.1758e-04	-0.00516
NLR	4.3569e-09	6.6007e-05	0.68322
ANN	2.7451e-07	5.2394e-04	-18.959
Max. Pressure (MPa)			
GPR	2.9301e+15	5.4130e+07	0.76064
SVM	5.6014e+15	7.4842e+07	0.54242
DT	3.1219e+14	1.7669e+07	0.97450
LR	5.2839e+15	7.2690e+07	0.56835
NLR	1.3611e+15	3.6893e+07	0.88881
ANN	2.1663e+14	1.4718e+07	0.98230
Max. Stress (MPa)			
GPR	9.2128e+15	9.5983e+07	0.83022
SVM	1.7201e+16	1.3115e+08	0.68301
DT	9.2517e+15	9.6186e+07	0.82950
LR	1.7213e+16	1.3120e+08	0.68278
NLR	9.1992e+15	9.5913e+07	0.83047
ANN	1.0965e+17	3.3114e+08	-1.02070

margin-based regressions are insufficient to model the complex tribological and geometric interactions inherent in the wheel-rail contact mechanics.

In summary, the methodology adopted in this study involved a systematic approach to model selection, training, and evaluation, leveraging a diverse set of regression techniques to address the complex problem of predicting wear in rail-wheel interactions. The outcomes of this study are expected to contribute significantly to the advancement of predictive maintenance strategies in railway systems.

Figure 9 shows the general flowchart of the predictive ML methodology.

To enhance the predictive performance of our models, we employed Bayesian optimization for hyperparameter tuning. Bayesian optimization is a probabilistic model-based optimization technique, ideal for optimizing complex functions that are expensive to evaluate. This method uses a Gaussian Process (GP) to model the function and applies the Bayes' theorem to update the model with each iteration, efficiently exploring the hyperparameter space to find the optimal settings.

The Bayesian optimization process significantly improved our models' accuracy by fine-tuning the hyperparameters based on the evaluation of a surrogate model of the objective function. This approach was particularly effective in optimizing parameters such as learning rate, regularization terms, and kernel parameters, which are crucial for the performance of models like SVM, ANN, and GPR.

The performance of each regression model, post hyperparameter optimization, was meticulously evaluated on the validation set. Figures 10, 11, 12, 13, 14 and 15 present surface plots that illustrate examples of the real and predicted data for the different regression algorithms. These visual representations provide an intuitive comparison between the models' predictions and the actual values, offering insights into the accuracy and reliability of each method.

Among the regression techniques employed, Linear Regression (LR) exhibited suboptimal performance, struggling to capture the complex non-linear relationships inherent in the rail-wheel interaction data. This limitation underscores the challenges of applying linear models to problems that exhibit high levels of non-linearity and interaction effects among input variables.

Conversely, the other regression models demonstrated commendable predictive capabilities, with Artificial Neural Networks (ANN) and Decision Trees (DT) emerging as the standout performers. Both ANN and DT models exhibited superior accuracy and generalization ability compared to their counterparts. The success of these models can be attributed to their flexibility in modeling non-linear relationships and their capacity to handle complex interaction

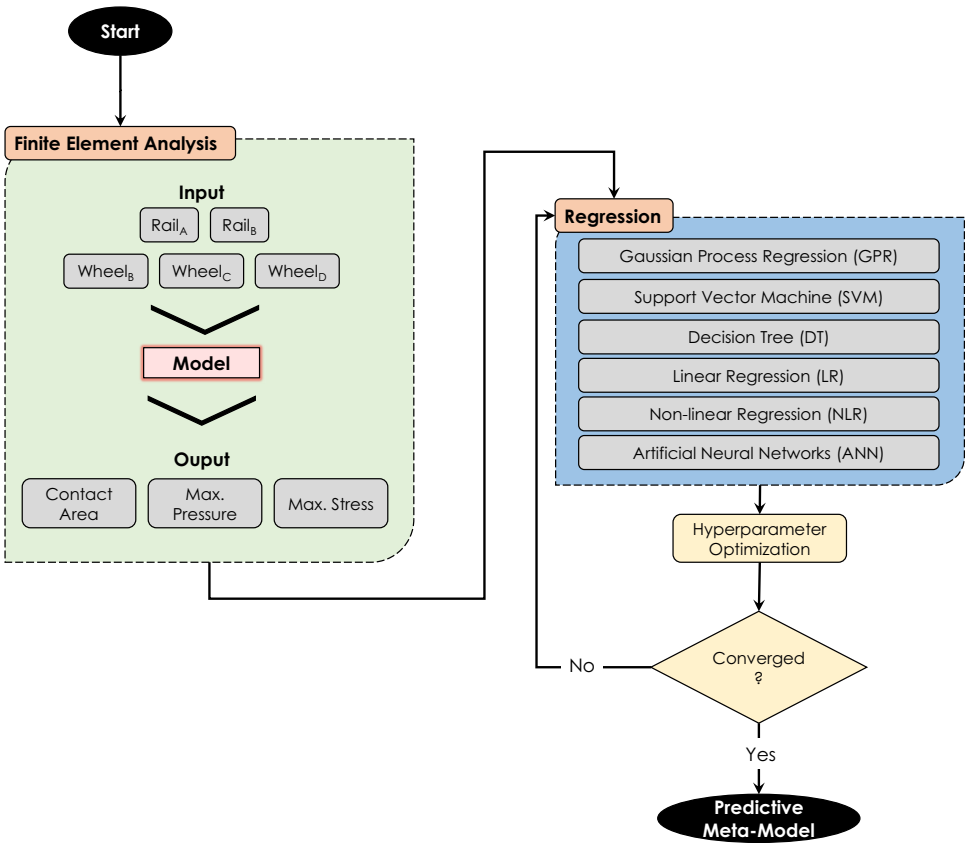


Fig. 9 General flowchart of the proposed predictive meta-model

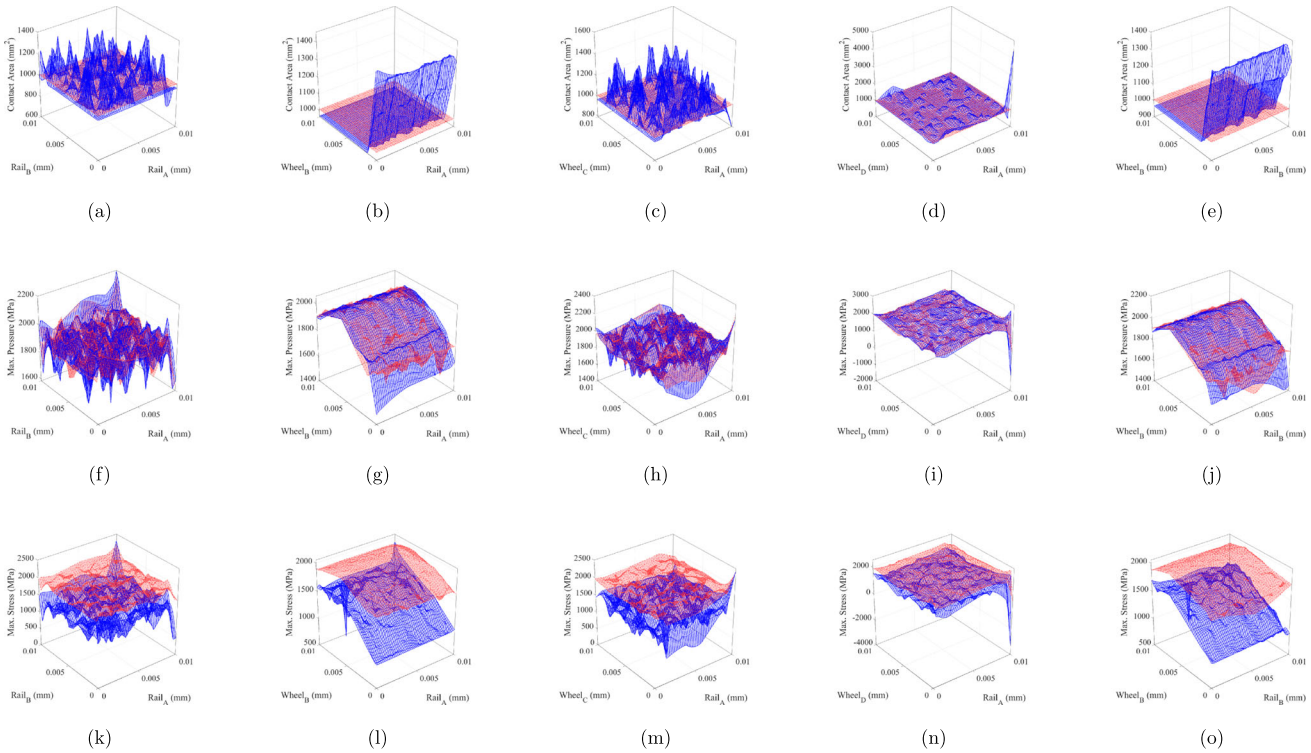


Fig. 10 Surface plot regression results for the Gaussian Process Regression (legend: ■ real and ■ predicted)

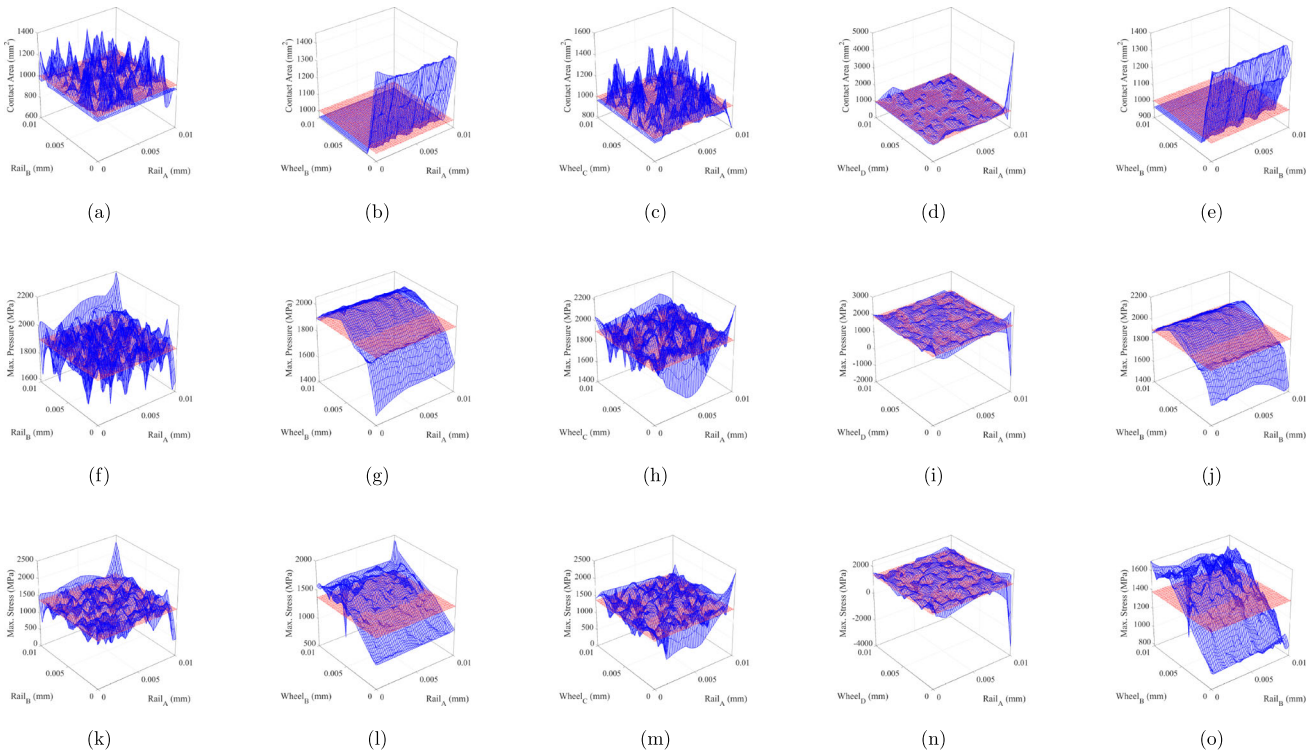


Fig. 11 Surface plot regression results for the Support Vector Machines (legend: ■ real and ■ predicted)

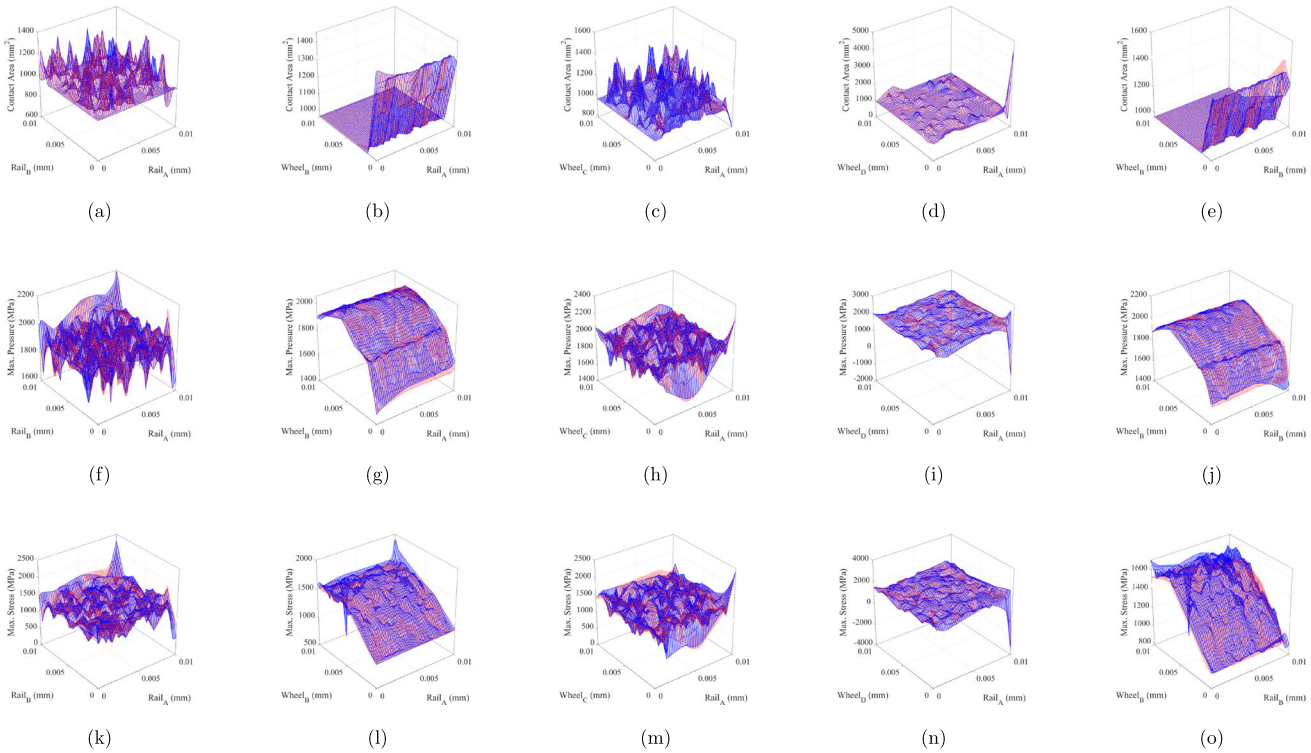


Fig. 12 Surface plot regression results for the Decision Tree (legend: ■ real and ■ predicted)

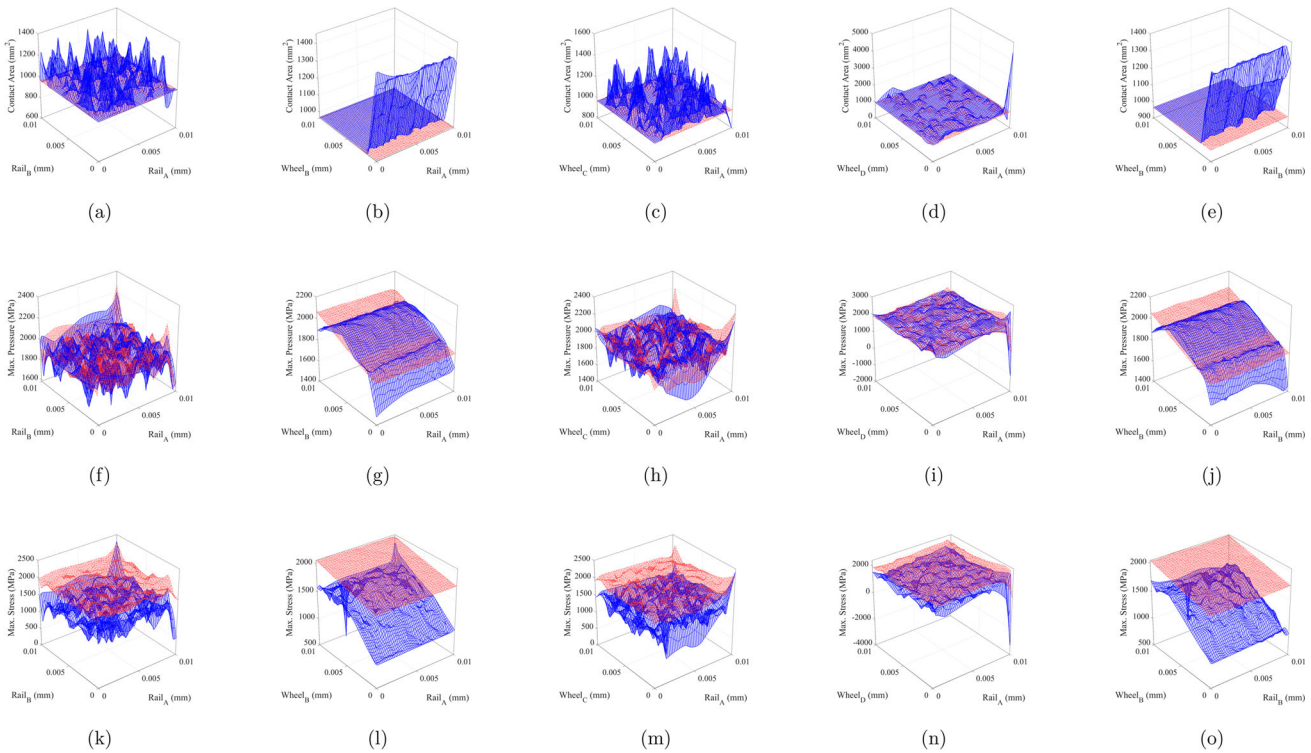


Fig. 13 Surface plot regression results for the Linear Regression (legend: ■ real and ■ predicted)

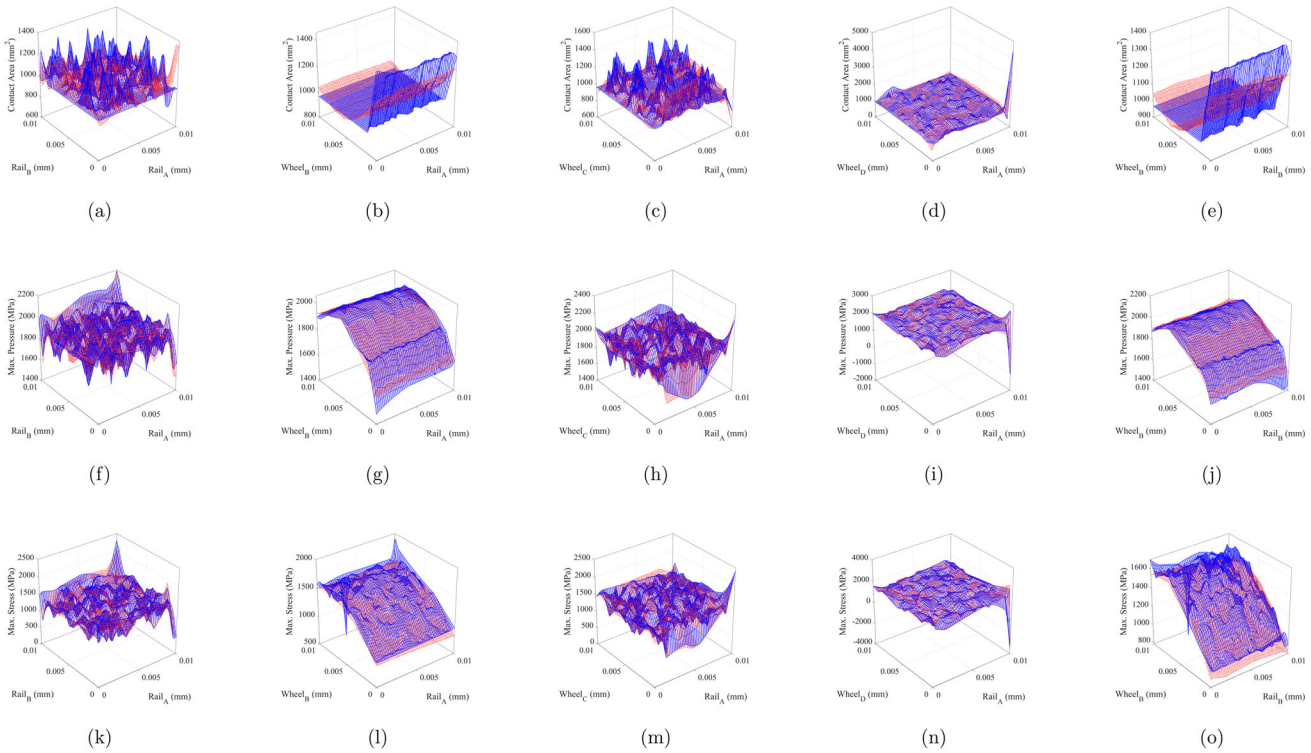


Fig. 14 Surface plot regression results for the Non-linear Regression (legend: ■ real and ■ predicted)

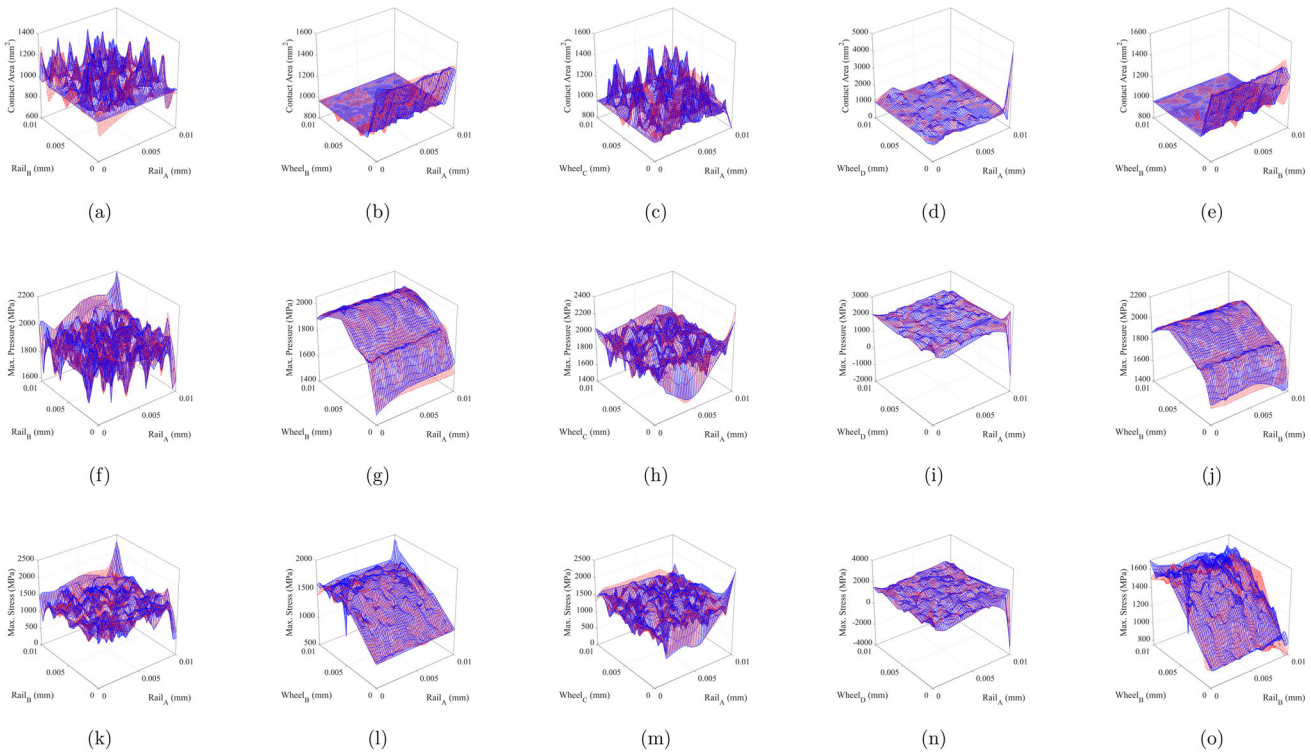


Fig. 15 Surface plot regression results for the artificial neural networks (legend: ■ real and ■ predicted)

patterns without stringent assumptions about the underlying data distribution.

The effectiveness of ANN and DT, as evidenced by their predictive performance, underscores their potential as robust and reliable tools for predicting wear in rail-wheel interactions. Their ability to discern intricate patterns and relationships in the data makes them particularly suited for applications in railway engineering, where accurate predictions can significantly contribute to maintenance planning and operational efficiency.

The comparative analysis of multiple regression techniques, supplemented by advanced hyperparameter optimization strategies, has yielded valuable insights into the modeling of wear phenomena in rail-wheel interactions. The findings from this study highlight the importance of selecting appropriate modeling approaches based on the characteristics of the problem at hand, with a particular emphasis on the advantages of non-linear models like ANN and DT in capturing complex system behaviors.

The surrogate model developed in this study serves a critical purpose by providing a computationally efficient alternative to running detailed Finite Element (FE) simulations for every possible wheel-rail configuration and condition. While FE models offer high-fidelity stress and contact mechanics data, they are computationally expensive and not suitable for real-time applications or scenarios requiring large-scale analysis. The surrogate model, based on machine learning, is trained on the outputs of the FE simulations and allows us to approximate key wear-related parameters with a fraction of the computational cost. This enables rapid evaluation of multiple geometries and loading conditions, making it a valuable tool for real-time predictive maintenance strategies in railway systems. By incorporating the surrogate model, we aim to bridge the gap between high-accuracy FE simulations and practical, scalable applications in railway maintenance and optimization.

4 Conclusion

In this study, we explored the predictability of wear in rail-wheel interactions using an integrated machine learning approach that encompassed advanced regression techniques, including Linear Regression, Non-linear Regression, Support Vector Machines, Decision Trees, Gaussian Process Regression, and Artificial Neural Networks. The validation of the developed Finite Element model demonstrated accuracy in simulating the complex dynamics of rail-wheel interaction, providing a reliable foundation for generating data for predictive analyses.

The comparison between regression techniques revealed significant differences in their ability to accurately predict wear, with Artificial Neural Networks and Decision Trees standing out for their superior performance. These models showed remarkable flexibility in modeling the non-linear and complex relationships present in the data, suggesting their robust applicability in real-world railway engineering scenarios.

Hyperparameter optimization through Bayesian Optimization techniques proved to be crucial for enhancing the predictive models' accuracy. This process allowed fine-tuning of the models, ensuring they operate at their optimal capacity, a critical step for the effective application of these techniques in predictive maintenance strategies within the railway sector.

Furthermore, the comparative analysis provided valuable insights into the strengths and limitations of each regression technique, emphasizing the importance of selecting the correct model based on the specific characteristics of the wear problem in rail-wheel interaction. The findings of this study underline the potential of machine learning approaches in predicting and managing wear, offering a valuable tool for optimizing maintenance and operational efficiency of railway systems.

While the proposed predictive models have demonstrated success in simulating wear under specific wheel-rail contact conditions, further validation is needed across a broader range of real-world scenarios, including varying rail materials, track geometries, and environmental factors. Future research should focus on full-scale experiments, dynamic testing, and integration with real-time railway monitoring systems. Specifically, a highly promising avenue for future empirical work involves installing accelerometers in vehicle bogies to continuously record dynamic vibrational responses over time. Pairing these real-time dynamic recordings with periodic in-situ measurements of rail and wheel wear would yield an outstanding, high-fidelity dataset. Such data would be invaluable for training and refining advanced machine learning methods, bridging the gap between numerical surrogate models and physical digital twins to ultimately optimize predictive maintenance, reduce downtime, and improve operational efficiency across various transportation infrastructures, such as tramways and high-speed trains.

In conclusion, this work contributes significantly to the field of railway engineering, providing a detailed and comparative framework for selecting appropriate predictive techniques. The insights gained are expected to guide future research and maintenance practices, promoting safer and more efficient railway systems through the adoption of advanced predictive maintenance strategies.

Acknowledgements The authors would like to acknowledge the financial support from CNPq (Conselho Nacional de Desenvolvimento Científico e Tecnológico) and FAPEMIG (Fundação de Amparo à Pesquisa do Estado de Minas Gerais).

Author Contributions K.B. developed the finite element model in ANSYS APDL, implemented the wear parametrization methodology, performed the numerical simulations, and contributed to the writing of the original draft. G.F.G. conceptualized and supervised the research, contributed to the development and implementation of the machine learning models, performed the regression analysis and hyperparameter optimization, and revised the manuscript critically for important intellectual content. D.A.G. contributed to the validation methodology, data interpretation, and critical revision of the manuscript. All authors reviewed, discussed the results, and approved the final version of the manuscript.

Funding The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614).

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

Competing interests The authors declare no competing interests.

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